202401471_Lab04 ICS

Dhirubhai Ambani University
Technology

Formerly DA-IICT

Introduction to Communication System[CT216].

I,Khushi Shah,Student Id: 202401471,declare that:

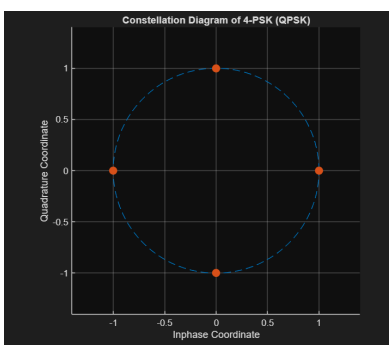
- The work that I am presenting is my own work and I have not copied the work that someone else has done.
- The concepts, understanding, and insights I will be describing are my own understood during ongoing classes.
- Wherever I have relied on an existing work from a source, I have provided a proper reference citation.
- I make this pledge truthfully. I know that violation of this solemn pledge can carry consequences.

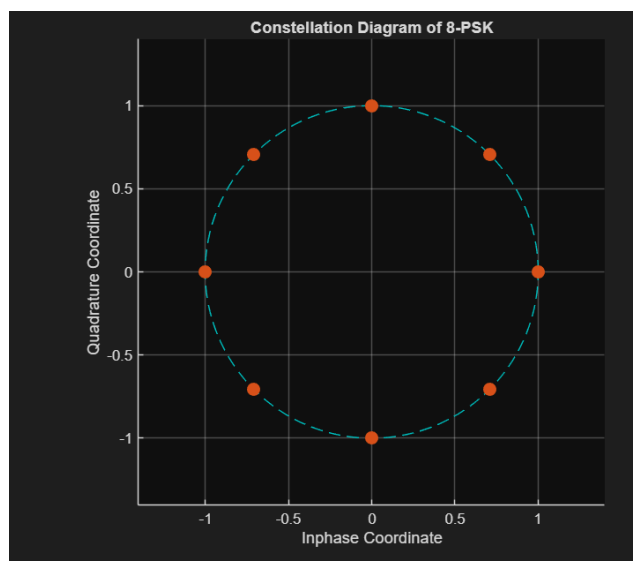
1. Modulator:

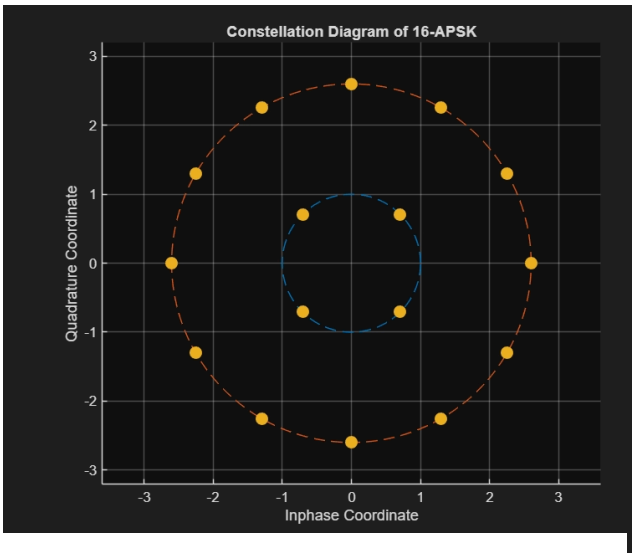
```
th = linspace(0,2*pi,500);

figure; hold on; grid on; axis equal;
plot(cos(th),sin(th),'--','Color',[0 0.45 0.74]);
scatter(S_qpsk(1,:),S_qpsk(2,:),80,[0.85 0.33 0.10],'fille
d');
xlabel('Inphase Coordinate'); ylabel('Quadrature Coordinat
e'); title(titles{1});
xlim([-1.4 1.4]); ylim([-1.4 1.4]);

figure; hold on; grid on; axis equal;
plot(cos(th),sin(th),'--','Color',[0 0.75 0.75]);
scatter(S_8psk(1,:),S_8psk(2,:),80,[0.85 0.33 0.10],'fille
d');
xlabel('Inphase Coordinate'); ylabel('Quadrature Coordinat
e'); title(titles{2});
xlim([-1.4 1.4]); ylim([-1.4 1.4]);

figure; hold on; grid on; axis equal;
plot(r1*cos(th),r1*sin(th),'--','Color',[0 0.45 0.74]);
plot(r2*cos(th),r2*sin(th),'--','Color',[0.85 0.33 0.10]);
scatter(S_16apsk(1,:),S_16apsk(2,:),80,[0.93 0.69 0.13],'fill
ed');
xlabel('Inphase Coordinate'); ylabel('Quadrature Coordinat
e'); title(titles{3});
xlim([-3.6 3.6]); ylim([-3.2 3.2]);

figure; hold on; grid on; axis equal;
cc={[0.93 0.69 0.13],[0 1 1],[0.56 0 1],[1 1 0],[0 0.75 1]};
radii=[0.5,1.5,2.0,2.5,3.0];
for r=1:5, plot(radii(r)*cos(th),radii(r)*sin(th),'--','Colo
r',cc{r}); end
scatter(S_32qam(1,:),S_32qam(2,:),70,[0 0.45 0.74],'filled');
xlabel('Inphase Coordinate'); ylabel('Quadrature Coordinat
```

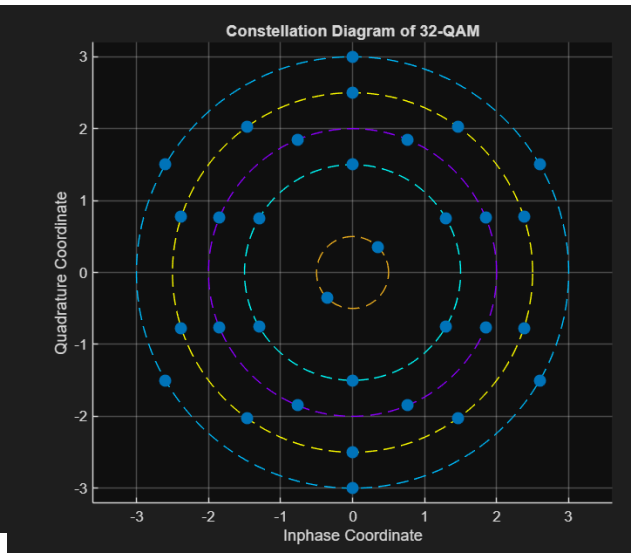
```
e'); title(titles{4});
xlim([-3.6 3.6]); ylim([-3.2 3.2]);

% Q1: Symbol Energy and Energy Per Bit
Eb_all = zeros(1,4);
for s = 1:4
Esym      = sum(S_all{s}.^2, 1);
Es        = mean(Esym);
Eb_all(s) = Es / k_all(s);
fprintf('%s: Es_bar=%.4f, Eb=%.4f\n', names{s}, Es, Eb_all
(s));
end

% Q1: Generate k*L Bits, Map to Symbols, Convert to Complex
S_cpx  = cell(1,4);
sl_all = cell(1,4);
isym_all = cell(1,4);

for s = 1:4
k        = k_all(s);
bits     = randi([0 1],1,k*L);                          % k*L
Bernoulli(0.5) bits
blocks   = reshape(bits,k,L);                           % L p
arallel k-bit blocks
isym     = blocks'*(2.^(k-1:-1:0))'+1;
isym     = isym(:)';                                    % 1xL
symbol indices
sl_all{s}   = S_all{s}(:,isym);                        % 2xL
transmitted symbol vectors
isym_all{s} = isym;
S_cpx{s}    = sl_all{s}(1,:) + 1j*sl_all{s}(2,:);     % compl
ex baseband symbols
end

% Q1: Squared Magnitude Spectrum (baseband, centred at 0 Hz)
figure;
```

```
for s = 1:4
NFFT = 1024;
mag2 = abs(fftshift(fft(S_cpx{s},NFFT))).^2;
f    = (-NFFT/2:NFFT/2-1)/NFFT;
subplot(2,2,s);
plot(f,10*log10(mag2+1e-12),'b'); grid on;
xlabel('Normalised Frequency'); ylabel('|S(f)|^2 (dB)'); titl
e(names{s});
end
sgtitle('Baseband Magnitude Spectra');
---------
%output is
QPSK (k=2): Es_bar=1.0000, Eb=0.5000
8-PSK (k=3): Es_bar=1.0000, Eb=0.3333
16-APSK (k=4): Es_bar=5.3200, Eb=1.3300
32-QAM (k=5): Es_bar=5.0781, Eb=1.0156
```

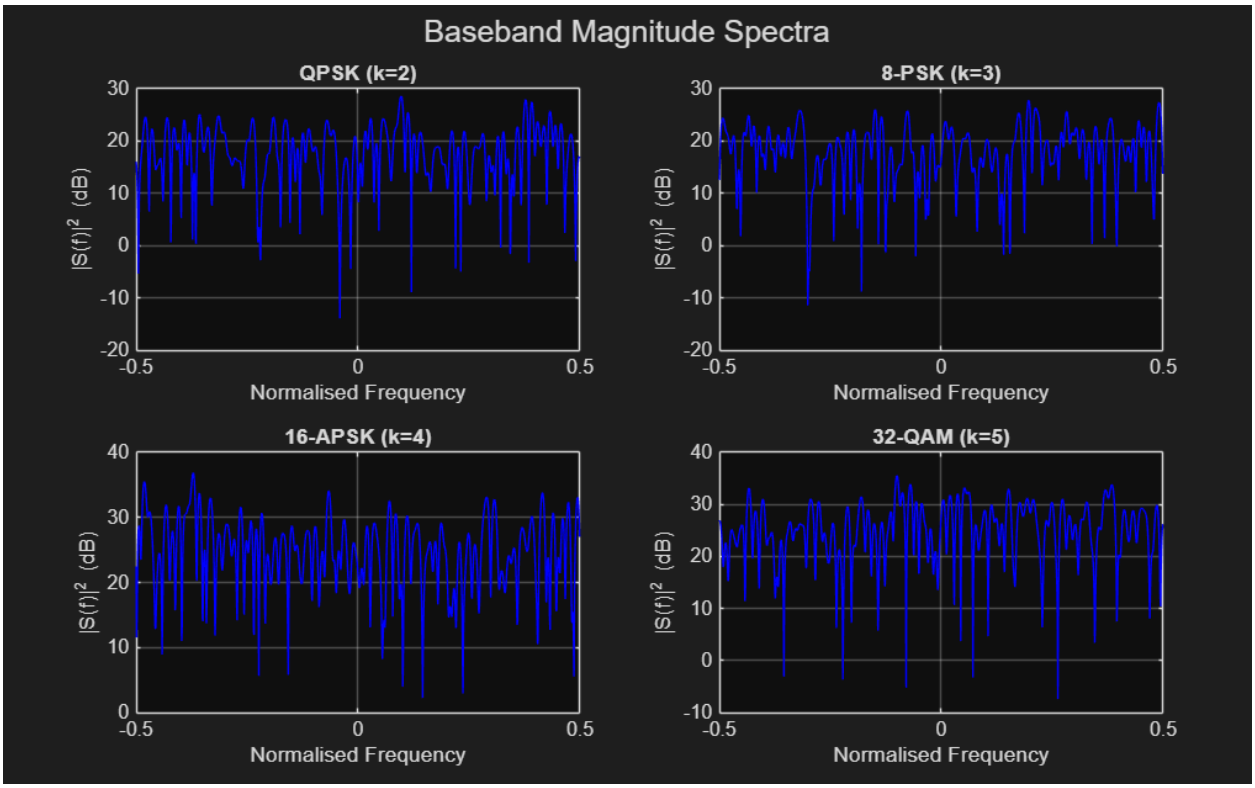








Q1 9 : Baseband magnitude spectra



2. Additive White Gaussian Noise (AWGN) channel:

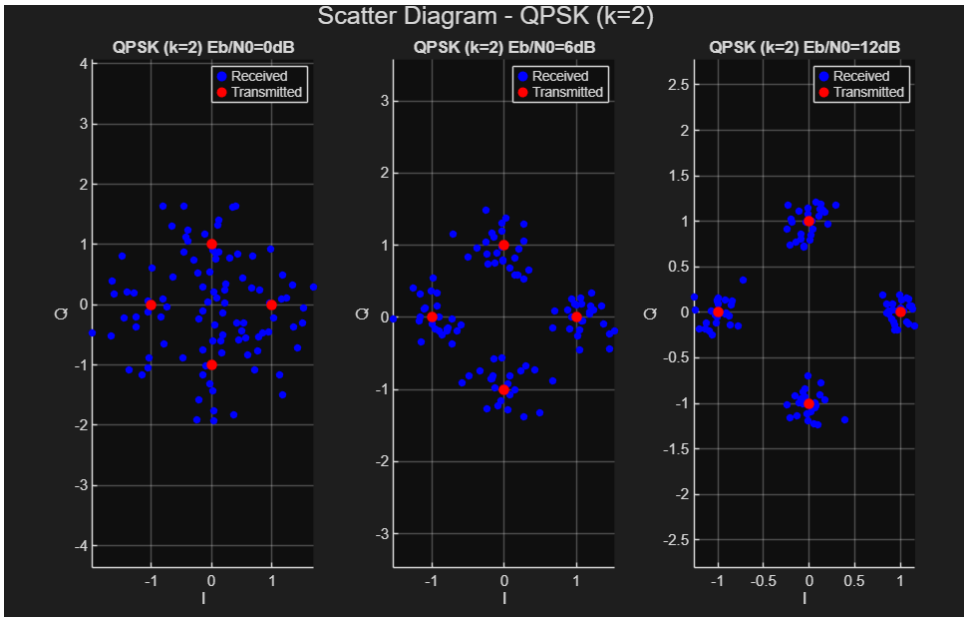
```
% --- Q2: AWGN Channel ---
`EbN0_dB  = 10;
EbN0_lin = 10^(0.1*EbN0_dB);`                                %
dB to linear

nl_all = cell(1,4);
for s = 1:4
sig2      = 0.5*Eb_all(s)/EbN0_lin;                    % sig^
2 = 0.5*Eb/(Eb/N0)lin
sig       = sqrt(sig2);
ni        = sig*randn(1,L);                             % I-c
hannel noise
nq        = sig*randn(1,L);                            % Q-cha
nnel noise
fprintf('%s: req sig^2=%.4f, var(ni)=%.4f, var(nq)=%.4f\n',na
mes{s},sig2,var(ni),var(nq));
nl_all{s} = [ni; nq];                          % 2xL noise ve
ctors nl=[ni_l; nq_l]
end
----------
%output is
QPSK (k=2): req sig^2=0.0250, var(ni)=0.0240, var(nq)=0.0232
8-PSK (k=3): req sig^2=0.0167, var(ni)=0.0147, var(nq)=0.0147
16-APSK (k=4): req sig^2=0.0665, var(ni)=0.0634, var(nq)=0.04
27
32-QAM (k=5): req sig^2=0.0508, var(ni)=0.0503, var(nq)=0.065
8
```

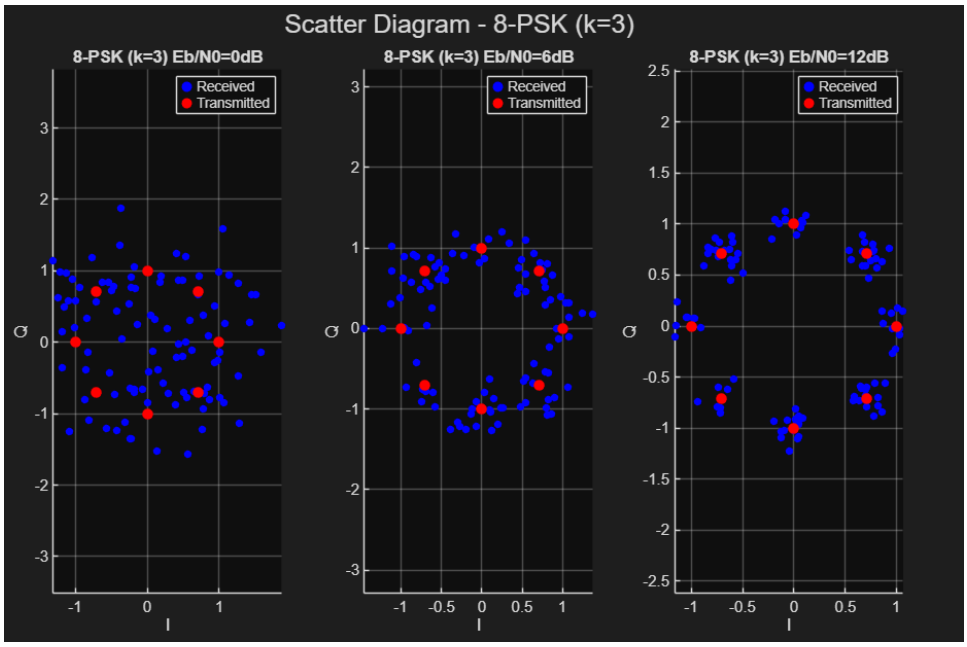
3. Demodulator:

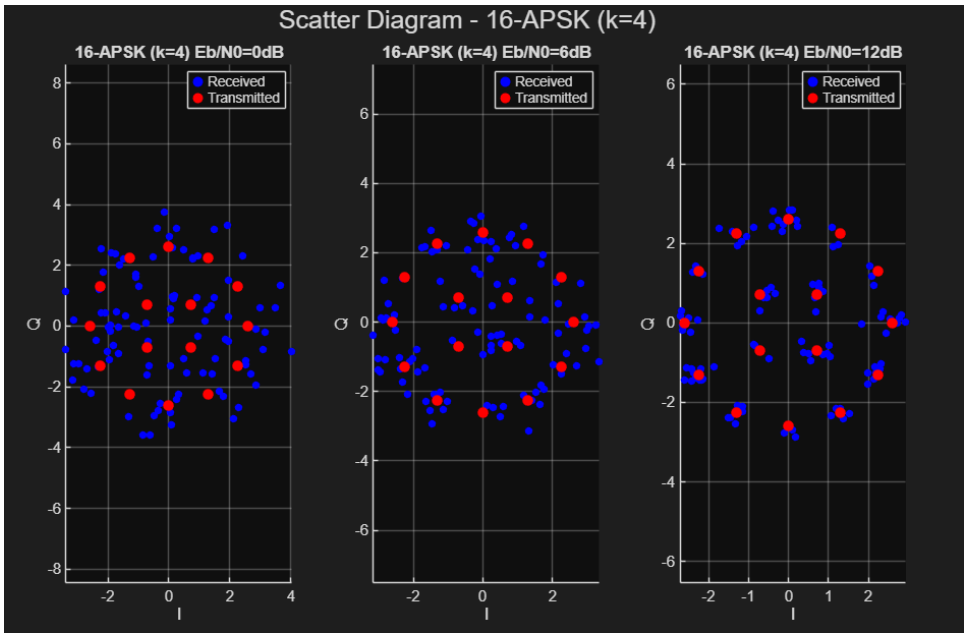
```
EbN0_dB_vals = [0, 6, 12];   % scatter at 0, 6, 12 dB

for s = 1:4
S  = S_all{s};
```

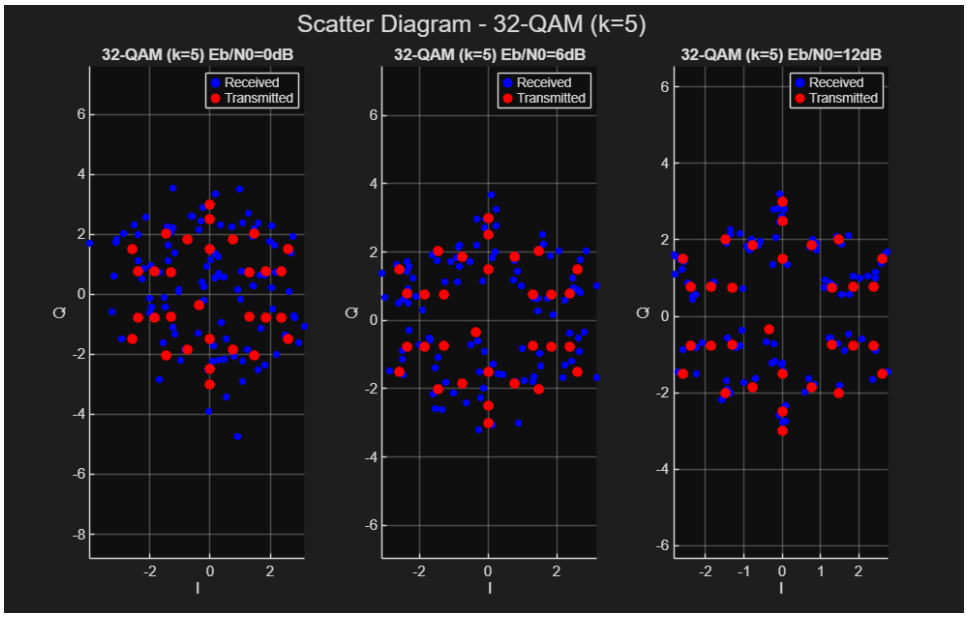
```
sl = sl_all{s};
Eb = Eb_all(s);

figure;
for v = 1:3
    sig_v = sqrt(0.5*Eb / 10^(0.1*EbN0_dB_vals(v)));
    rl_v  = sl + [sig_v*randn(1,L); sig_v*randn(1,L)]; % rl =
sl + nl
    subplot(1,3,v); hold on; grid on; axis equal;
    scatter(rl_v(1,:),rl_v(2,:),20,'b','filled');       % rec
eived
    scatter(sl(1,:),sl(2,:),40,'r','filled');            % tr
ansmitted
    xlabel('I'); ylabel('Q');
    title(sprintf('%s Eb/N0=%ddB',names{s},EbN0_dB_vals(v)));
    legend('Received','Transmitted','Location','best');
end
sgtitle(['Scatter Diagram - ',names{s}]);

% min-distance detection with noise disabled: verify 0 errors
detected = zeros(1,L);
for l = 1:L
    [~,detected(l)] = min(sum((S - sl(:,l)).^2, 1));   % clos
est constellation point
end
fprintf('%s (no noise): errors = %d/%d\\n',names{s},sum(detec
ted ~= isym_all{s}),L);
end
----------
%output is
QPSK (k=2) (no noise): errors = 0/100
8-PSK (k=3) (no noise): errors = 0/100
16-APSK (k=4) (no noise): errors = 0/100
32-QAM (k=5) (no noise): errors = 0/100
```

Scatter Diagrams:


Scatter Diagram - QPSK (k=2)
QPSK (k=2) Eb/N0=0dB
QPSK (k=2) Eb/N0=6dB
QPSK (k=2) Eb/N0=12dB
Received
Transmitted
Q
I


Scatter Diagram - 8-PSK (k=3)
8-PSK (k=3) Eb/N0=0dB
8-PSK (k=3) Eb/N0=6dB
8-PSK (k=3) Eb/N0=12dB
Received
Transmitted
Q
I



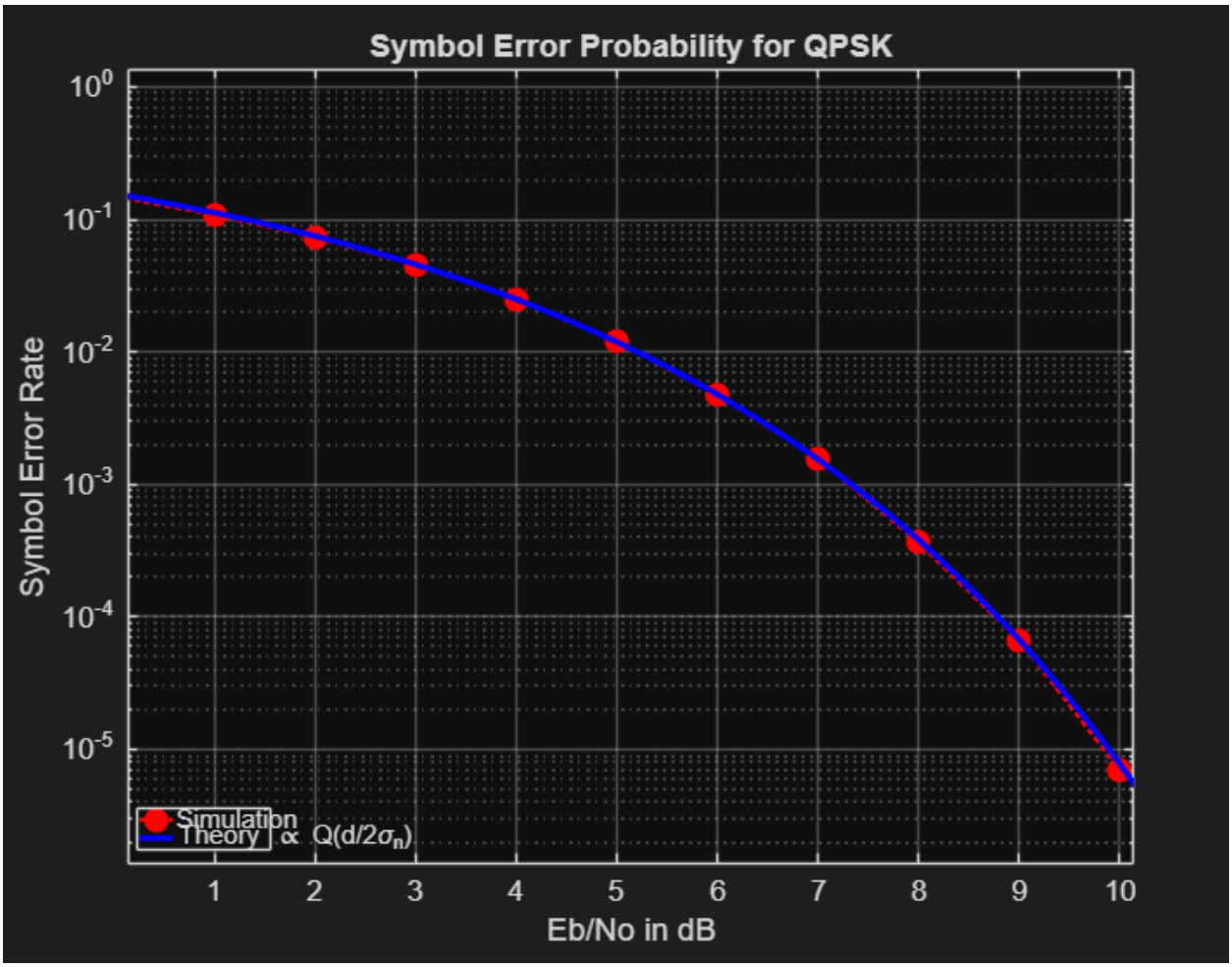


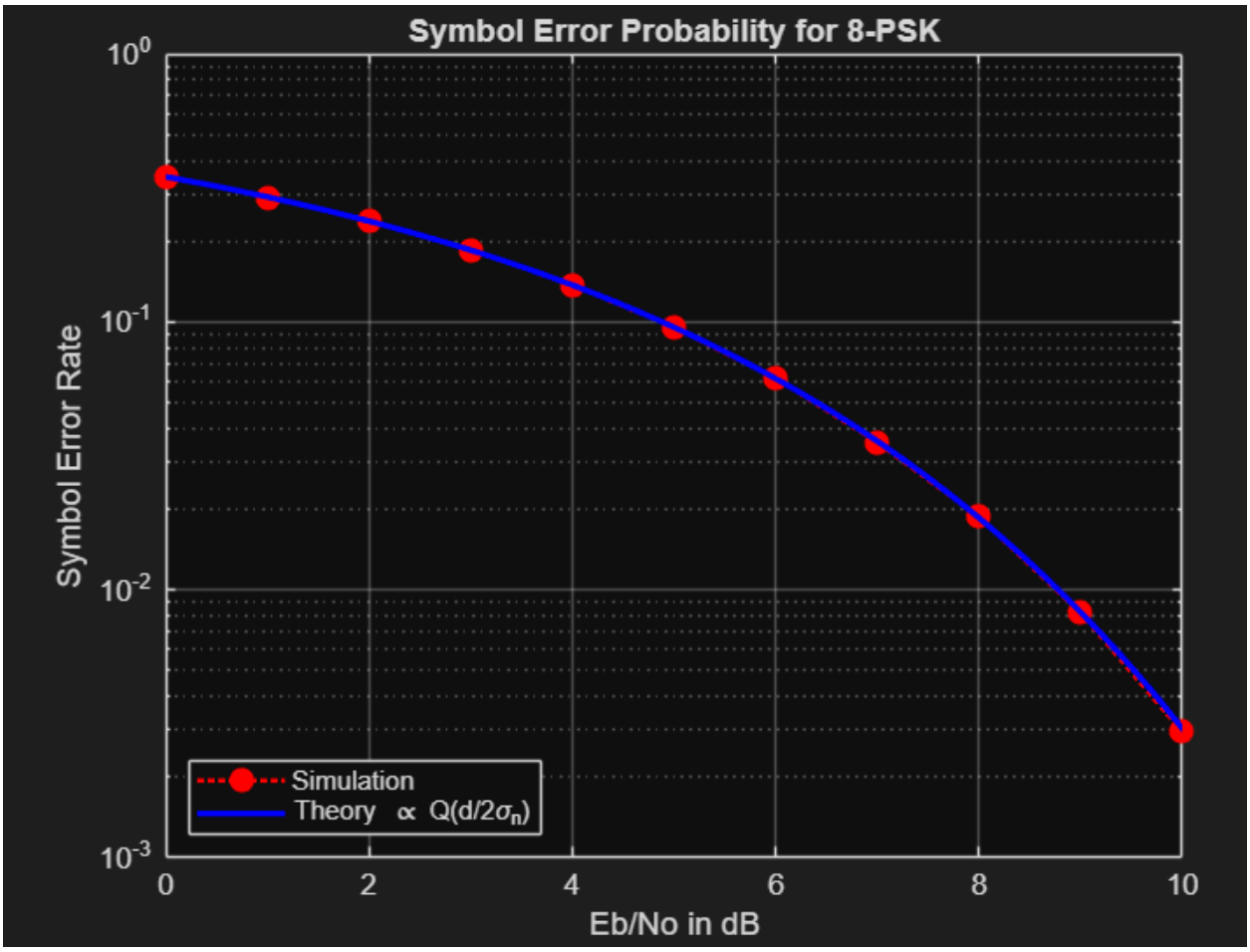
4. Monte-Carlo Simulator:

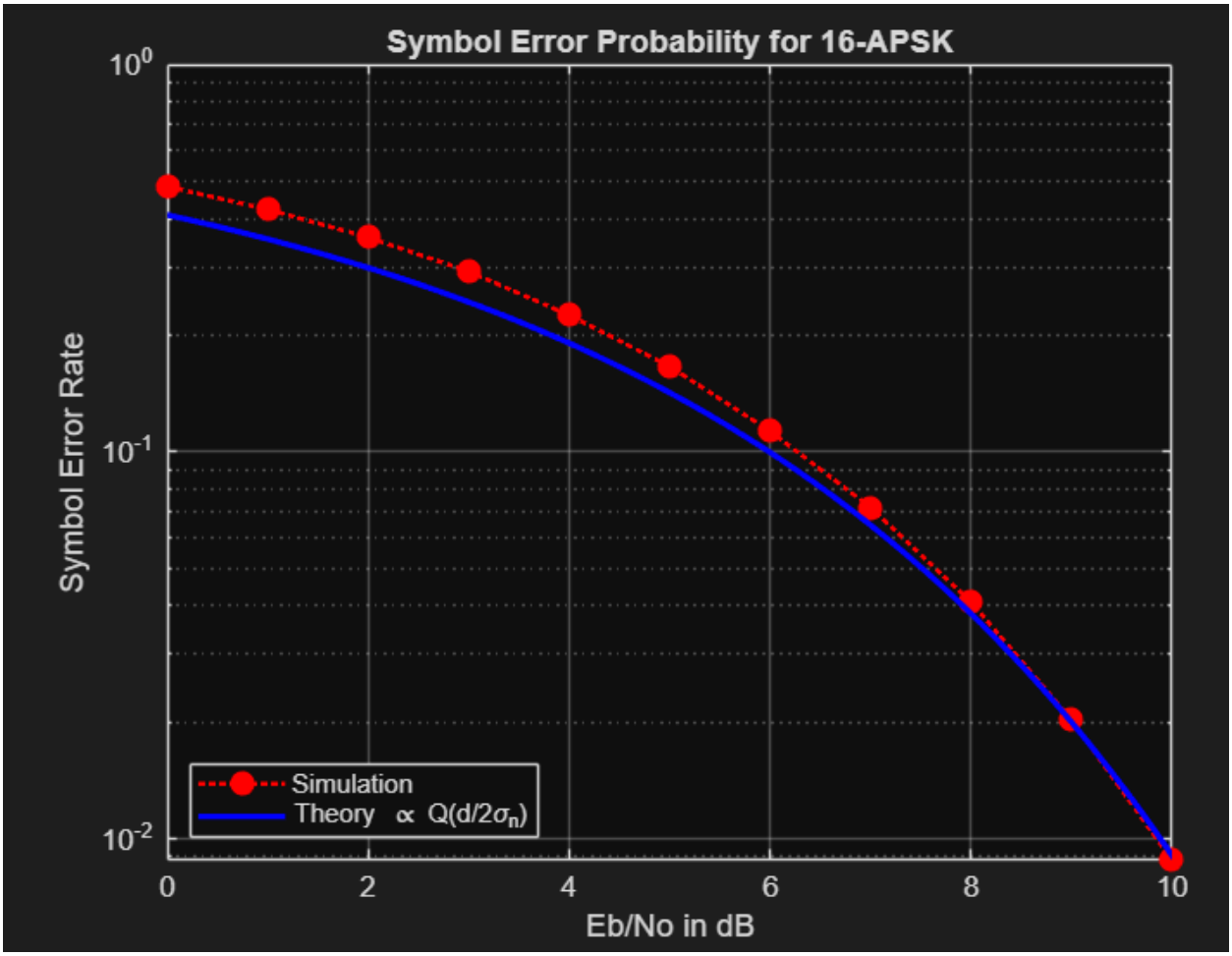
```
EbN0_sim_vec = 0:1:10;                                  % simulat
ion: 0 to 10 dB, 1 dB steps
EbN0_th_vec  = 0:0.1:15;                          % theory: 0 to 15
```

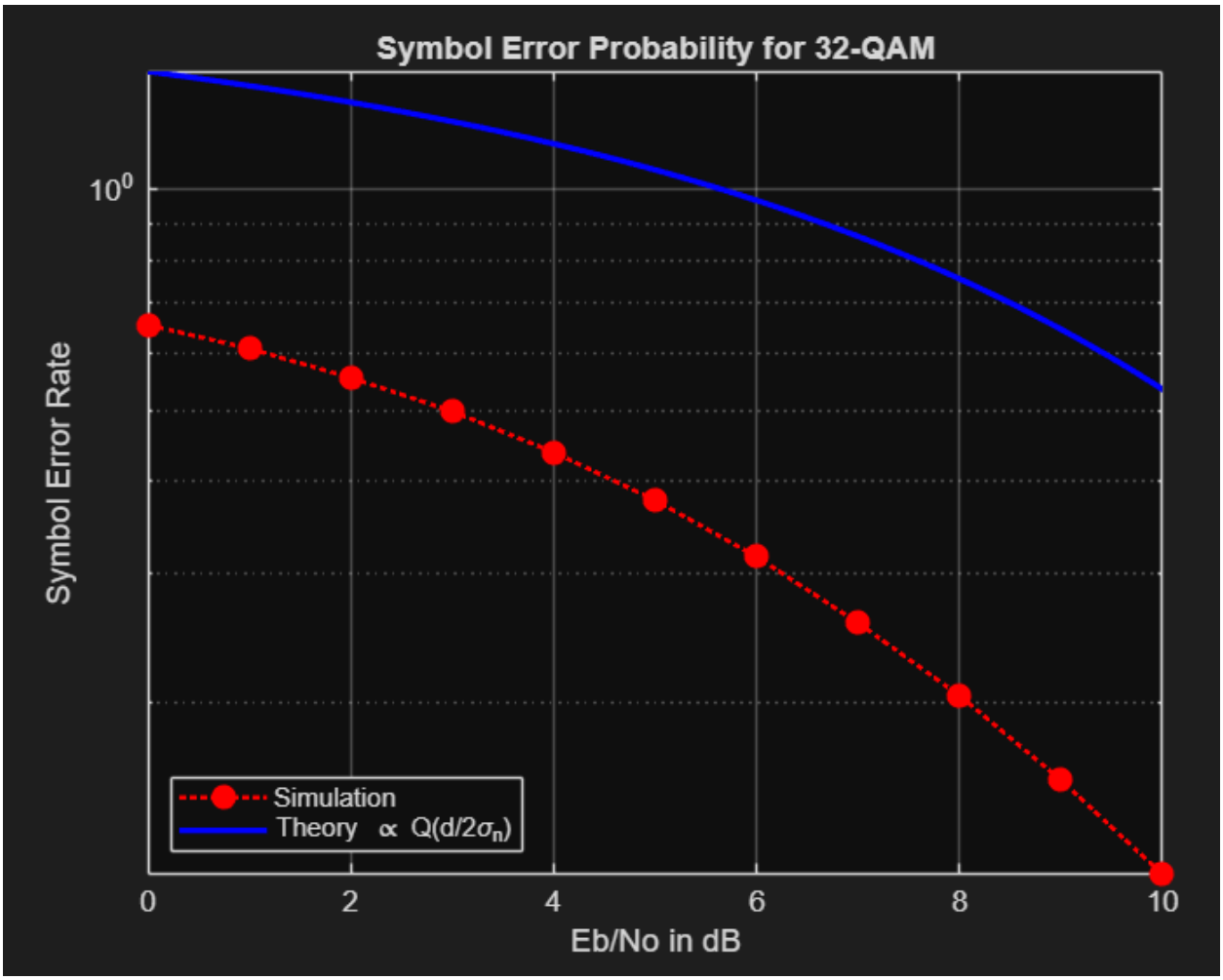
```
dB, 0.1 dB steps (smooth)
Nneighbor    = [2, 2, 2, 4];       % nearest neighbours (spe
c: 2 for PSK/APSK, 4 for QAM)
plot_titles  = {'Symbol Error Probability for QPSK',  'Symbol
Error Probability for 8-PSK','Symbol Error Probability for 16
-APSK','Symbol Error Probability for 32-QAM'};

for s = 1:4
S  = S_all{s};
k  = k_all(s);
M  = M_all(s);
Eb = Eb_all(s);
Nn = Nneighbor(s);

d_min = Inf;
for m1 = 1:M
    for m2 = m1+1:M
        d_min = min(d_min, sqrt(sum((S(:,m1)-S(:,m2)).^2)));
    end
end

Psym_theory = zeros(1,length(EbN0_th_vec));
for ii = 1:length(EbN0_th_vec)
    N0_i            = Eb / 10^(0.1*EbN0_th_vec(ii));   % N0 =
Eb/(Eb/N0)
    Psym_theory(ii) = Nn * 0.5*erfc(sqrt(d_min^2/(2*N0_i))/sq
rt(2));
end

% Monte-Carlo simulation
Psym_sim = zeros(1,length(EbN0_sim_vec));
for ii = 1:length(EbN0_sim_vec)
    sig_i   = sqrt(0.5*Eb / 10^(0.1*EbN0_sim_vec(ii)));
    err_sum = 0;
    for t = 1:Nsim
        bits_t   = randi([0 1],1,k*L);
```

```
        blocks_t = reshape(bits_t,k,L);
        isym_t   = blocks_t'*(2.^(k-1:-1:0))'+1;
        isym_t   = isym_t(:)';
        sl_t     = S(:,isym_t);
        rl_t     = sl_t + [sig_i*randn(1,L); sig_i*randn(1,
L)]; % add AWGN
        det_t    = zeros(1,L);
        for l = 1:L
            [~,det_t(l)] = min(sum((S - rl_t(:,l)).^2,1)); %
min-distance detect
        end
        err_sum = err_sum + sum(det_t ~= isym_t);
    end
    Psym_sim(ii) = err_sum/(Nsim*L);      % average SER over
Nsim trials
end

figure('Position',[100 100 600 450]);
semilogy(EbN0_sim_vec, Psym_sim,'r:o','LineWidth',1.5,'Marker
FaceColor','r','MarkerSize',7); hold on;
semilogy(EbN0_th_vec,  Psym_theory, 'b-',  'LineWidth',2);
grid on;
legend('Simulation','Theory \\propto Q(d/2\\sigma_n)','Locati
on','southwest');
title(plot_titles{s},'FontWeight','bold');
ylabel('Symbol Error Rate'); xlabel('Eb/No in dB');
xlim([0 10]);
end
```


Symbol Error Probability for QPSK
Symbol Error Rate
Eb/No in dB
Simulation
Theory ∝ Q(d/2σn)
10^0
10^-1
10^-2
10^-3
10^-4
10^-5
1
2
3
4
5
6
7
8
9
10


Symbol Error Probability for 8-PSK
Symbol Error Rate
Eb/No in dB
10^0
10^{-1}
10^{-2}
10^{-3}
0
2
4
6
8
10
Simulation
Theory ∝ Q(d/2σn)


Symbol Error Probability for 16-APSK
Symbol Error Rate
Eb/No in dB
Simulation
Theory ∝ Q(d/2σn)
10^0
10^-1
10^-2
0
2
4
6
8
10



Attaching the full .m file along with this pdf.